

Student Name:

Reg/ID #:

Q1: Assume a processor has a base CPI of 1.5 and a clock rate of 1GHz. The main memory access time is 200 ns. Suppose the instruction miss rate at L1 cache is 2%. Calculate CPI for the processor with L1 cache and compute speedup compared to the ideal processor i.e. no misses.

Sol:

$$\text{CPI}_{\text{new}} = \text{CPI}_{\text{base}} + \text{CPI}_{\text{mem}}$$

$$\text{CPI}_{\text{mem}} = I * \text{miss_rate} * \text{miss_penalty}$$

$$\text{miss_rate} = 2\% = 2/100$$

$$\text{miss_penalty} = 200 \text{ nsec (Needed in clock cycles)}$$

$$\text{miss_penalty} = 200 / (1/1\text{GHz}) = 200 \text{ cc}$$

$$\text{CPI}_{\text{mem}} = I * 200 * (2/100) = 4I$$

$$\text{CPI}_{\text{new}} = 1.5 + 4 = 5.5 \text{ (could drop } I \text{ here)}$$

$$\text{Speedup} = \text{CPI}_{\text{base}} / \text{CPI}_{\text{mem}} = 1.5 / 5.5 = 0.2727 \text{ (relative to ideal processor)}$$

Q2: Now assume if we add a L2 cache to the processor in Q1 that has a 20 clock cycles average access time (AMAT i.e. for both miss and hit) and reduces the miss rate to main memory to 0.5%, calculate speedup for the processor (compared to only L1 cache) having both L1 and L2 cache.

Sol:

$$\text{CPI}_{\text{new}} = \text{CPI}_{\text{base}} + \text{CPI}_{\text{mem}}$$

$$\text{CPI}_{\text{mem}} = \text{CPI}_{\text{L1}} + \text{CPI}_{\text{L2}}$$

For L1: 2% misses in total. 0.5% in main memory and 1.5% (2%-0.5%) in L2 cache.

$$\text{CPI}_{\text{L1}} = I * (1.5/100) * 20 = 0.3I \text{ (20 is used directly since it is given as clock cycles and NOT time)}$$

For L2: 0.5% misses in total, however, each miss uses access time for L2 plus the access time of main memory

$$\text{CPI}_{\text{L2}} = I * (0.5/100) * (20 + 200) = 1.1I$$

$$\text{CPI}_{\text{mem}} = \text{CPI}_{\text{L1}} + \text{CPI}_{\text{L2}} = 0.3I + 1.1I = 1.4I$$

$$\text{CPI}_{\text{new}} = \text{CPI}_{\text{base}} + \text{CPI}_{\text{mem}} = 1.4 + 1.5 = 2.9 \text{ (could drop } I \text{ here)}$$

$$\text{Speedup} = 5.5 / 2.9 = 1.89 \text{ (relative to L1)}$$

$$\text{Speedup} = 1.5 / 2.9 = 0.517 \text{ (relative to ideal processor)}$$